# Test Plan: Inventory Management System (IMS) for Thrifted & Branded Surplus Clothing (with Item Approval Workflow)

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## 1. Introduction

This Test Plan defines how the IMS for Thrifted & Branded Surplus Clothing will be verified, including its scope, quality objectives, team responsibilities, test approach, and completion criteria. Because the physical inspection of garments happens outside the system, testing focuses on whether the system correctly records the inspection result and enforces the status rules that depend on it.

### 1.1 Scope

#### 1.1.1 In Scope

- Login and role-based access control for Staff, Reviewer, and Admin.
- Adding new items (brand, size, condition, price, photo) with an automatic "Pending" status.
- Input validation on the Add Item form and the Sale screen.
- The Pending Items queue and the Reviewer's Approve and Reject actions.
- The mandatory rejection reason and its storage against the item.
- Status transitions and the Status Guard Rule (Pending → In Stock → Sold, with rejection returning to Pending).
- Sales processing for "In Stock" items only.
- Stock tracking and low-stock flagging.
- Inventory search and the dashboard.
- Status reports and the status change log.
- Non-functional checks on usability (click count), performance (search load time), and security (RBAC, hashed passwords).

#### 1.1.2 Out of Scope

- Point-of-sale (POS) hardware such as barcode scanners, receipt printers, and cash drawers.
- External marketplace synchronization, such as Shopee and Lazada.
- RFID scanning or tagging.
- A customer-facing storefront.
- AI-based authenticity detection; authenticity is judged by the Reviewer through physical inspection.

### 1.2 Quality Objectives

- Verify that every user can perform only the actions permitted by their role.
- Verify that every new item is saved with the status "Pending" and that required item fields are validated.
- Verify that Approve moves an item from "Pending" to "In Stock".
- Verify that Reject returns an item to "Pending" and always stores a rejection reason.

- Verify that no item can skip a status, and that only "In Stock" items can be sold.
- Verify that stock counts, low-stock flags, search results, and reports are accurate and consistent with the status log.
- Verify that every status change is logged with the user, previous status, new status, and timestamp.
- Verify that the non-functional requirements are met: 2 clicks or fewer to approve or reject, search results in under 2 seconds, and passwords stored as hashes.

### 1.3 Roles and Responsibilities

| Role | Responsibilities |
|---|---|
| Project Manager | Oversees the test schedule, tracks progress against the Gantt chart, coordinates the team, and approves the final sign-off. |
| System Developer | Builds the system, fixes reported defects, supports the test environment and test data, and re-delivers builds for retesting. |
| UI/UX Designer | Provides the approved layouts used as the reference for interface checks and corrects usability issues found in testing. |
| QA | Defines the test strategy and quality criteria, reviews test cases and results, manages defect triage, and verifies exit criteria. |
| System Analyst | Maintains requirements traceability, clarifies expected behavior, and leads feedback gathering and usability evaluation during acceptance testing. |
| Tester | Writes and executes test cases, runs the Postman collections, logs defects, and performs retesting and regression testing. |

## 2. Test Methodology

### 2.1 Overview

Testing uses a black-box functional approach. Testers verify the system against the documented requirements by providing inputs through the user interface and the API and comparing the outputs and resulting item statuses with the expected results, without examining the internal code. Each test case is traced to a functional or non-functional requirement. Status rules, API constraints, role-based access, and search are automated with Postman collections. Screen-level checks, such as the 2-click approve/reject requirement, and acceptance checks are performed manually.

### 2.2 Test Levels / Types

| Test Type | Description | Tool / Method |
|---|---|---|
| Functional | Verifies each function against its requirement, including add item, approve, reject, sale, search, | Postman collections and manual execution |

| Test Type | Description | Tool / Method |
|---|---|---|
| | and reports. | |
| Integration | Verifies that the modules work together, such as intake, approval, sales, stock counts, and the status log, and that API rules are enforced. | Postman collections |
| System | Verifies the complete system end to end in the test environment, including role-based flows and non-functional requirements. | Manual and automated end-to-end scenarios |
| Smoke | Quick check of each new build to confirm that login, add item, approve, and sale work before deeper testing begins. | Postman collections (run on every new build) |
| Regression | Re-runs the affected and core test cases after defect fixes or changes to confirm that nothing else is broken. | Postman collections |
| Acceptance Testing (role-based walkthrough) | Testers playing the Staff, Reviewer, and Admin roles confirm that the system supports the intended workflow and meets the requirements. | Manual, using scripted scenarios |

### 2.3 Initial Functional Test Scenarios

| ID | Area | Test Condition | Expected Result |
|---|---|---|---|
| IMS-01 | RBAC (Staff) | A Staff user attempts to use the Approve/Reject actions, open the reports screen, or open any Reviewer or Admin URL directly. | Access is denied and the user is redirected or shown an authorization error. |
| IMS-02 | RBAC (Reviewer and Admin) | A Reviewer attempts to add an item or process a sale, and an Admin attempts to approve or reject an item. | The actions are unavailable in the interface and rejected by the API. |
| IMS-03 | Add Item | Staff submits the Add Item form with a valid brand, size, condition, price, and photo. | The item is saved with the status "Pending" and appears in the Pending Items queue. |
| IMS-04 | Input Validation | Staff submits the form with a required field empty, a non-numeric or negative price, or an unsupported photo file. | The submission is blocked and a clear error message is shown; no item is created. |

| ID | Area | Test Condition | Expected Result |
|---|---|---|---|
| IMS-05 | Approve | A Reviewer selects "Approve" on a Pending item. | The status changes to "In Stock" and the item becomes available for sale. |
| IMS-06 | Reject | A Reviewer selects "Reject" on a Pending item and enters a rejection reason. | The item returns to "Pending" and the reason is saved and visible on the item record. |
| IMS-07 | Missing Rejection Reason | A Reviewer selects "Reject" and leaves the reason blank. | The action is blocked with a message that a reason is required; the status does not change. |
| IMS-08 | Status Skip Prevention | A user or API request attempts to sell a "Pending" item, or to set an item directly from "Pending" to "Sold". | The system refuses the action, the status remains "Pending", and an error message is returned. |
| IMS-09 | Sales Processing | Staff processes a sale for an "In Stock" item. | The status changes to "Sold", the stock count decreases by one, and the sale is recorded. |
| IMS-10 | Low-Stock Flag | The number of "In Stock" items in a category falls to or below the configured threshold of 5. | A low-stock alert is shown on the Admin dashboard for that category. |
| IMS-11 | Search | An Admin searches by brand, size, condition, and status. | Only matching items are returned, and a search with no matches shows an empty-result message. |
| IMS-12 | Reports | An Admin generates a status report for Pending, In Stock, and Sold items. | The report totals match the actual item statuses and include rejection reasons. |
| IMS-13 | Status Logging | A status change is made through approve, reject, or sale. | A log entry is created with the item, previous status, new status, user, and timestamp. |
| IMS-14 | Usability | A Reviewer approves or rejects an item starting from the Pending Items queue. | The action is completed in 2 clicks or fewer (plus typing the reason on rejection). |
| IMS-15 | Performance | An Admin runs an inventory search on a database of 2,000 or more items. | Results load in under 2 seconds. |

## 2.4 Bug Triage

| Severity | Definition | Action |
|---|---|---|
| Critical | The system crashes, data is lost or corrupted, or a core rule fails, such as a Pending item being sold or an authentication bypass. | Testing of the affected area stops; the developer fixes it immediately, and the fix is retested and regression tested before testing resumes. |
| High | A major function does not work as specified, such as approve, reject, or sale failing, with no acceptable workaround. | Fixed within the current test cycle and retested before the exit criteria are assessed. |
| Medium | A function works partially or incorrectly but a workaround exists, such as an inaccurate report total or a wrong validation message. | Scheduled for a fix in the current or next build, depending on schedule. |
| Low | A cosmetic or minor usability issue with no effect on function, such as alignment, spelling, or label wording. | Logged and fixed as time allows; may be deferred with QA approval. |

## 2.5 Acceptance Criteria

### 2.5.1 Entry Criteria

- The requirements and this Test Plan are approved by the Project Manager and QA.
- The system build is deployed to the test environment and passes the smoke test.
- The test environment, test database, and user accounts for Staff, Reviewer, and Admin are ready.
- Test cases are written and reviewed, and the Postman collections are available.

### 2.5.2 Exit Criteria

- 100% of planned test cases have been executed.
- At least 95% of executed test cases have passed.
- There are zero unresolved Critical or High defects.
- All Medium and Low defects are either fixed or documented with a QA-approved decision to defer.
- Regression testing after the final fixes has passed.
- The acceptance test results are recorded and accepted.

### 2.5.3 Suspension and Resumption

- Testing is suspended when a Critical defect blocks further testing, when the smoke test fails, or when the test environment or test database becomes unavailable.
- Testing resumes when the defect is fixed and verified, a new build passes the smoke test, and the environment is restored with the test data reset.

## 2.6 Test Completion / Sign-off

Testing is complete when the exit criteria in section 2.5.2 are met and the Test Summary Report is reviewed. Sign-off is given in writing by QA, the System Analyst, and the Project Manager, and the System Developer acknowledges that all agreed defects have been resolved. Any criterion that is not met must be listed as a known issue with its severity and an agreed decision.

## 2.7 Project Tasks and Draft Schedule

The schedule below follows the updated Gantt chart: system testing runs Oct 26–Nov 1 (WBS 5.1), acceptance testing Nov 2–8 (WBS 5.2), and defect fixing, regression, and deployment Nov 9–19 (WBS 6.1).

| Task | Responsible | Timing |
|---|---|---|
| Draft Test Plan | QA & Tester | September 11–13 |
| Document functional and non-functional requirements | System Analyst | September 16–20 |
| Write test cases and prepare test data | Tester & QA | September 28–October 25 |
| Develop Postman collections and test scripts | Tester | October 12–27 |
| Prepare and verify the test environment | System Developer & Tester | October 22–25 |
| Smoke and functional tests (Postman collections and manual) | Tester | October 28–29 |
| Integration and status-rule tests (Postman collections) | Tester | October 29–30 |
| Defect logging and system test report | Tester & QA | October 31–November 1 |
| Acceptance testing with testers in the Staff, Reviewer, and Admin roles | Tester & QA | November 2–4 |
| Gather feedback and evaluate usability | System Analyst | November 5–6 |
| Acceptance test report and sign-off | QA & System Analyst | November 7–8 |
| Defect fixing and refinement (from test and acceptance reports) | System Developer & Tester | November 9–13 |
| Regression and final testing | QA & Tester | November 14–16 |
| Test Summary Report and sign-off | QA & Project | November 17–19 |

| Task | Responsible | Timing |
|---|---|---|
| | Manager | |

## 3. Test Deliverables

### Before Testing

- Test Plan (this document)
- Requirements document and requirements traceability matrix
- Test cases and test scenarios, including IMS-01 to IMS-15
- Test data set, including at least 2,000 sample items and user accounts for each role
- Postman collections and test scripts

### During Testing

- Test execution logs and results for each cycle
- Defect reports with severity, steps to reproduce, and screenshots
- Postman Collection Runner reports
- Daily or weekly test progress reports
- Updated traceability matrix

### After Testing

- Test Summary Report with pass rate, defect counts by severity, and open issues
- Final defect log with status and resolution
- Regression test results
- Acceptance test results and tester feedback summary
- Test sign-off document

## 4. Resource and Environment Needs

### 4.1 Testing Resources

| Resource | Description |
|---|---|
| Test team | QA, Tester, System Analyst, System Developer, UI/UX Designer, and Project Manager, as defined in section 1.3. |
| Test workstations | Laptops or desktops running Windows 11 with Google Chrome and Microsoft Edge. |
| Test server and database | A dedicated test deployment of the IMS with its own test database, separate from development. |
| Automation tools | Postman collections for automated API tests; screen-level checks are run manually. |

| Resource | Description |
| --- | --- |
| Test data | Sample items across brands, sizes, and conditions in every status, a set of at least 2,000 items for performance and scalability checks, and accounts for Staff, Reviewer, and Admin. |
| Defect tracking and documentation | A shared tracker or spreadsheet for defects, test results, and the Gantt chart workbook for schedule tracking. |
| Acceptance testers | Two to three testers (for example classmates or teammates outside the build team) who play the Staff, Reviewer, and Admin roles and run the scripted acceptance scenarios. A real thrift or surplus seller may join if available. |

### 4.2 Test Environment

Testing is performed in a dedicated test environment that is separate from development so that unfinished work and test data do not affect each other. The environment runs on Windows 11 workstations and is accessed through the latest stable versions of Google Chrome and Microsoft Edge, which are the supported browsers for the web-based IMS.

The environment has its own test database, which is seeded with controlled sample data for every item status and for the Staff, Reviewer, and Admin accounts, and can be reset to a known state before each test cycle and after any suspension. Postman collections run against the deployed test application and verify status rules, the rejection reason requirement, role-based endpoint access, and search results and response time, while screen-level checks are performed manually in the supported browsers. Each build is deployed to this environment only after the developer confirms it is ready, and the smoke test must pass before formal testing begins.